

Gavin Bayley

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Third-year Mechatronics and Robotics Engineering student with hands-on experience in robotics research, embedded systems, and mechanical design/fabrication across multiple competition teams and a research lab.

EDUCATION

Queen's University

Kingston, ON

Bachelor of Applied Science in Mechatronics and Robotics Engineering

Sept. 2024 – Present (Third-Year Student)

- Charles Allan Thompson Summer Research Award 2026 – \$6,000 Value
- GPA: 4.07 / 4.30 – Dean's Scholar Distinction
- Queen's Chancellor's Scholarship - \$48 000 Value

Double Dogwood (Bilingual English/French) Diploma

Victoria, BC

Oak Bay High School

June 2024

- Clive B. Simmons Award (STEM) · BC Achievement Award · School District Authority Award for project-based learning

WORK EXPERIENCE

Engineering Contract Work

Expected Start – October 2026

MIthRIL Lab, Queen's University – Supervisor: Prof. Matthew Pan

Kingston, ON

- Incoming role assisting with the development of a robotic rollator for the Connected Minds CINTHeA project, supporting assistive-mobility robotics research

Robotics Systems Research Assistant, MIthRIL Lab

May 2026 – Aug. 2026

Queen's University – Supervisor: Prof. Matthew Pan

Kingston, ON

- Designed, fabricated, and assembled a tendon-driven, continuum-stem animatronic flower robot utilizing multi-modal fabrication as the physical platform for an LLM-driven mental health companion robot research project
- Built the full ROS 2 (Humble) host software stack: a Tkinter GUI for manual and kinematic control with a live 3D preview, a forward-kinematics solver converting bend targets into antagonistic servo commands, and a command-mux to connect to the robot over micro-ROS/Wi-Fi
- Developed embedded C++ firmware (PlatformIO, ESP32) driving 4 servos through an I2C controller, petal LEDs, and an encoder-controlled flower petal motor, with a self-healing Wi-Fi/micro-ROS reconnection protocol using mDNS discovery
- Implemented computer vision (OpenCV) for ball tracking and face detection, using HSV-segmentation ball localization and RetinaFace to lay the groundwork for closed-loop and attention-based behaviors
- Authored a 27-page handbook covering assembly, tuning, software architecture, and setup/operation to hand off a fully reproducible build to the lab

TECHNICAL EXPERIENCE & COMPETITION TEAMS

Mechanical Team Member

April 2026 – Present

F1Tenth – Ingenuity Labs Racing (autonomous 1/10-scale race car competition)

Kingston, ON

- 3D scanned the vehicle to produce an accurate 3D model, used to design and fabricate custom electronics mounts for the car

Project Manager

2025 – Present

Rellectric Car Team – Purdue EV Grand Prix (collegiate electric go-kart competition)

Kingston, ON

- Designed and led the build of a dedicated practice kart used for driver training

Simulation & Integration Team Member

2025 – 2026

Queen's AutoDrive Team – SAE AutoDrive Challenge II (Level 4 autonomous vehicle)

Kingston, ON

- Built a predictive lookahead algorithm that analyzes upcoming track geometry to reduce behavior-planner computation load
- Designed a watchdog framework that monitors the health of a 40+ node ROS 2 stack in real time

Mechanical Team Member

2024 – 2025

Queen's VEX U Robotics Team – International Collegiate VEX U League

Kingston, ON

- Prototyped the climbing mechanism for the 2025 VEX Robotics World Championship robot
- Designed and fabricated custom brackets and mounts for both the competition and worlds-qualifying robots

Vehicle Restoration

2022 – 2025

1973 Mercedes 450SL (full restoration) · 1976 Rolls-Royce Silver Shadow (mechanical restoration)

Victoria, BC

- Diagnosed and repaired electrical and mechanical systems, including high-pressure hydraulic braking and suspension

COACHING AND INSTRUCTING

Outdoor & Athletic Instructor

2020 – 2025

Multiple Employers

Victoria, BC

- Instructed and mentored diverse groups of students across outdoor, alpine, and aquatic settings, enforcing safety standards while communicating technical skills to learners of varying abilities
- Outdoor Education Instructor, Western Educational Adventures (2020–25) · Ski Instructor, Mount Washington Alpine Resort (2021–24) · Racing Ski Coach, Mount Washington Ski Club (2023–24) · Swim Instructor & Lifeguard, Oak Bay Recreation Centre (2023–24)

TECHNICAL SKILLS

Languages: C · C++ · Python · Arduino · Julia · VHDL · NIOS II Assembly

Robotics & Embedded: ROS 2 · micro-ROS · ESP32/PlatformIO firmware · I2C · OpenCV

CAD & Fabrication: Autodesk Fusion 360 · SolidWorks · OnShape · 3D Printing · Laser Cutting

Machine Shop: CNC Milling · CNC Lathe Operation · Manual Mill/Lathe · Welding

Hardware & Electronics: Microcontrollers · Circuit Assembly · Soldering · Mechanical Diagnostics

Tools: Git · Tkinter · Foxglove Studio · LATEX · Microsoft Suite